DRAFT SCHEMATIC / PROPOSED DESIGN

FastVCL: proposed sparse-to-tube workflow

1
Fixed frame features
Query / reference
ISC descriptors
M x d and N x d

→

2
Sparse correspondences
Sample feature indices
Tiled mutual Top-K
Retain index mapping

→

3
Temporal proposals
Robust track fitting
Check support on both axes
Split unsupported gaps

→

4
Local refinement
Bounded tube regions
Monotone path alignment
Recover interval boundaries

→

5
Copy intervals
Validate and deduplicate
Zero or more segment pairs
Attach ranking scores

Scope
Given video pairs; fixed pretrained descriptors

Training claim
No additional downstream model training

Open design choices
Sampling, alignment, and scoring require specification and validation.

Conceptual architecture. The prototype is incomplete; arrows do not certify implementation or effectiveness.